

Task 1 - Part 2: Evaluation Strategy for Text-to-SQL Systems

Overview

The core question is: how do we know if a Text-to-SQL agent is actually working well? This section outlines a simple evaluation framework that will be used in Tasks 3 and 4.

Evaluation Dimensions

1. Execution Success

Does the SQL run without errors? If the query throws a syntax error or references a wrong column, it fails completely. This is the minimum bar.

2. Result Accuracy

Does the query return the correct data? This matters more than whether the SQL looks identical to the reference. Two different queries can produce the same correct result.

3. Exact Match

Does the generated SQL match the reference SQL exactly? This is stricter. Useful but not always fair since there are multiple valid ways to write the same query.

4. Table and Column Selection

Did the agent pick the right tables and columns? An agent might write valid SQL but join the wrong tables entirely.

5. Join Correctness

Did the agent use the right join type? Using INNER JOIN when LEFT JOIN is needed changes results significantly.

6. Aggregation Correctness

For questions involving counts, sums, or averages, did the agent use the right function and group correctly?

7. Handling Ambiguous Questions

Some questions like "show recent orders" are vague. Does the agent make a reasonable assumption or break completely?

8. Self-Correction

If the first query fails, can the agent retry and fix itself? Good agents recover from errors.

Scoring

Metric	Weight
Result set accuracy	30%
Execution success	25%
Structural correctness	20%
Schema linking	15%
Self-correction	10%

Connection to Tasks 3 and 4

The 50 ground truth queries from Part 1 will be used as the test set. The agent built in Tasks 3 and 4 will receive each natural language question, generate SQL, execute it, and the output will be compared against the reference queries using the scoring approach above.